

Name:

YATIN PREETAM BHOSWANI

Roll No:

241211

• Write your name cleanly in **CAPITAL** letters and roll number inside the boxes.

• Write answers in the space provided only.

Maximum marks: 25

Time: 18:30 - 19:45

1. (a) Find an example of a space M and a strict contraction map $F : M \rightarrow M$ having no fixed point where M is not complete.(b) Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a map defined for any $(x_1, x_2) \in \mathbb{R}^2$ by $F(x_1, x_2) = \frac{1}{4}(x_1 + x_2, x_1 + 2x_2)$. Show that F is a strict contraction and find the unique fixed point. [2+4]

(a) ~~$M = [0, 1]$~~ $M = (0, 1)$
is an incomplete space
and $f(x) = \frac{x}{2}$.

also $|f(y) - f(x)| = \left| \frac{y}{2} - \frac{x}{2} \right|$
 $\leq \frac{1}{2} |y - x|$

(a) take $M = (0, 1)$

this is incomplete space
($x_n : \frac{1}{n}$ is Cauchy but does not converge)

let $F : M \rightarrow M$
 $F(x) = \frac{x}{2}$

show F is strict contraction

$$|F(y) - F(x)| = \left| \frac{y}{2} - \frac{x}{2} \right| = \frac{1}{2} |y - x| \leq \frac{1}{2} |y - x|$$

$$\Rightarrow |F(y) - F(x)| \leq \frac{1}{2} |y - x|$$

eqn: $x = \frac{x}{2}$ has no roots in $(0, 1)$

(b) $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$F(x_1, x_2) = \frac{1}{4}(x_1 + x_2, x_1 + 2x_2)$$

take $y, x \in \mathbb{R}^2$

$$y = (y_1, y_2), \quad x = (x_1, x_2)$$

to show it is

$$\|F(y, y_2) - F(x_1, x_2)\|$$

$$= \left\| \frac{1}{4}(y_1 + y_2, y_1 + 2y_2) - \frac{1}{4}(x_1 + x_2, x_1 + 2x_2) \right\|$$

to show it is contraction

$$d(y, x) \quad d(F(y), F(x)) \leq \alpha d(y, x)$$

here d is the $\|\cdot\|$ norm

$$\left\| \frac{1}{4}(y_1 + y_2, y_1 + 2y_2) - \frac{1}{4}(x_1 + x_2, x_1 + 2x_2) \right\|$$

$$= \frac{1}{4} \|(y_1 + y_2 - x_1 - x_2, y_1 + 2y_2 - x_1 - 2x_2)\|$$

$$= \frac{1}{4} \sqrt{(y_1 + y_2 - x_1 - x_2)^2 + (y_1 + 2y_2 - x_1 - 2x_2)^2}$$

$$= \frac{1}{4} \sqrt{y_1^2 + y_2^2}$$

$$= \frac{1}{4} \sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2 + 2(y_1 - x_1)(y_2 - x_2) + (y_1 - x_1)^2 + (2y_2 - 2x_2)^2 + 2(y_1 - x_1)(2y_2 - 2x_2)}$$

$$= \frac{1}{4} \sqrt{2(y_1 - x_1)^2 + 5(y_2 - x_2)^2 + 6(y_1 - x_1)(y_2 - x_2)}$$

$$= \frac{1}{4} \sqrt{(y_1 - x_1)^2 + \frac{5}{2}(y_2 - x_2)^2 + 3(y_1 - x_1)(y_2 - x_2)}$$

$$= \frac{1}{4} \sqrt{\frac{1}{16}}$$

$$= \sqrt{\frac{1}{8}(y_1 - x_1)^2 + \frac{5}{16}(y_2 - x_2)^2 + \frac{6}{16}(y_1 - x_1)(y_2 - x_2)}$$

$$\leq \sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2} = \|y - x\|$$

(as $\frac{1}{8}(y_1 - x_1)^2 + \frac{5}{16}(y_2 - x_2)^2 + \frac{6}{16}(y_1 - x_1)(y_2 - x_2) < (y_1 - x_1)^2 + (y_2 - x_2)^2$)

$\Rightarrow \|f(y) - f(x)\| < \|y - x\|$
hence a strict contraction.

finding fixed pt. (unique)

$$F(x_1, x_2) = (x_1, x_2)$$

$$\frac{1}{4}(x_1 + x_2, x_1 + 2x_2) = (x_1, x_2)$$

$$\begin{cases} x_1 = x_1 + x_2 \\ x_2 = x_1 + 2x_2 \end{cases} \Rightarrow \begin{cases} x_1 = 0 \\ x_2 = 0 \end{cases}$$

fixed pt is $(0, 0)$

..... Rough Work

fixed pt. strict contraction mapping & complete space.

if $\|f(x) - f(y)\| < \alpha d(x, y)$

if i take x_0

$(f^n(x_0))$ is Cauchy

$$x \Rightarrow \lim_{n \rightarrow \infty} f^n(x_0)$$

$$\frac{1}{4}(x_1 + x_2, x_1 + 2x_2) = (x_1, x_2)$$

$$\frac{1}{4}(x_1 + 2x_2) = x_2$$

$$x_1 + 2x_2 = 4x_2$$

$$x_1 = 2x_2$$

$$2[y_1^2 + x_1^2 - 2y_1x_1]$$

$$+ 5[y_2^2 + x_2^2 - 2y_2x_2]$$

$$+ 6y_1y_2 - 6y_1x_2$$

$$- 6x_1y_2 + 6x_1x_2$$

$$2y_1^2 + 2x_1^2 - 4y_1x_1$$

$$+ 5y_2^2 + 5x_2^2 - 10y_2x_2$$

$$+ 6y_1y_2 - 6y_1x_2$$

$$- 6x_1y_2 + 6x_1x_2$$

$$2y_1^2 + 5y_2^2 + 2x_1^2$$

$$+ 5x_2^2 - 10y_1x_2$$

2. (a) Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be uniformly continuous, and define $f_n(x) := g(x - \frac{1}{n^2})$. Show that f_n uniformly converges to g on $\mathbb{R}$.

(b) Let $f: [0, 3] \rightarrow \mathbb{R}$ be a continuous function. Justify whether $\lim_{n \rightarrow \infty} \int_{\frac{1}{n}}^{2+\frac{1}{n}} f(y) dy = \int_0^2 f(y) dy$.

(a) $g: \mathbb{R} \rightarrow \mathbb{R}$ is uniformly continuous.

$$\forall \epsilon > 0, \exists \delta_\epsilon > 0 \text{ s.t. } |y - x| < \delta_\epsilon \Rightarrow |g(y) - g(x)| < \epsilon \quad \forall x, y \in \mathbb{R}.$$

to show $f_n(x) := g(x - \frac{1}{n^2})$ uniformly converges to g

i.e. $\forall \epsilon > 0, \exists N_\epsilon \text{ s.t. } |f_n(x) - g(x)| < \epsilon \quad \forall x, n > N_\epsilon$

Soln: take $\epsilon > 0$.

$$\exists \delta_\epsilon \text{ s.t. } |y - x| < \delta_\epsilon \Rightarrow |g(y) - g(x)| < \epsilon \quad \forall x, y \in \mathbb{R}.$$

$$y \in (x - \delta_\epsilon, x + \delta_\epsilon) \Rightarrow |g(y) - g(x)| < \epsilon \quad \forall x, y \in \mathbb{R}.$$

take $\delta_\epsilon = \frac{1}{\sqrt{\epsilon}}$ (sq. root)

$$\Rightarrow y \in (x - \frac{1}{N_\epsilon^2}, x + \frac{1}{N_\epsilon^2})$$

$$\Rightarrow |g(y) - g(x)| < \epsilon$$

we know $x - \frac{1}{n^2} > x - \frac{1}{N_\epsilon^2}$

$$\forall n > N_\epsilon$$

and that $x - \frac{1}{n^2} < x + \frac{1}{N_\epsilon^2}$

$$\Rightarrow x - \frac{1}{n^2} \in (x - \frac{1}{N_\epsilon^2}, x + \frac{1}{N_\epsilon^2})$$

$$\forall n > \frac{1}{\sqrt{\epsilon}} N_\epsilon$$

$$\Rightarrow |g(x - \frac{1}{n^2}) - g(x)| < \epsilon$$

$$\forall n > \frac{1}{\sqrt{\epsilon}} N_\epsilon$$

$$\& \forall x.$$

take $N = N_\epsilon + 1$ (since it is $n > N_\epsilon$)

$$\Rightarrow |g(x - \frac{1}{n^2}) - g(x)| < \epsilon \quad \forall x, n > N$$

$$\Rightarrow |f_n(x) - g(x)| < \epsilon \quad \forall n, n > N$$

hence $f_n(x)$ converges uniformly to $g(x)$.

(b) $\lim_{n \rightarrow \infty} \int_{\frac{1}{n}}^{2+\frac{1}{n}} f(y) dy$ as f is continuous integral exists.

By Fundamental Theorem of Calculus. (Riemann integral)

$$\int_a^b f(y) dy = F(b) - F(a)$$

also this $F(x)$ is continuous and differentiable.

$$\int_{\frac{1}{n}}^{2+\frac{1}{n}} f(y) dy = F(2+\frac{1}{n}) - F(\frac{1}{n})$$

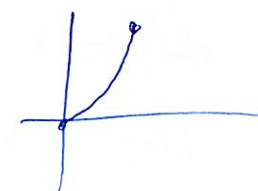
$$\lim_{n \rightarrow \infty} F(2+\frac{1}{n}) - F(\frac{1}{n}) = F(\lim_{n \rightarrow \infty} 2+\frac{1}{n}) - F(\lim_{n \rightarrow \infty} \frac{1}{n})$$

(as F is continuous)

$$= F(2) - F(0)$$

$$= \int_0^2 f(y) dy$$

..... Rough Work



$$\begin{aligned} \text{let } \int_a^b f(x) dx &= F(a) - F(b) \\ &= \lim_{n \rightarrow \infty} (F(\frac{2}{n}) - F(\frac{1}{n})) \end{aligned}$$

continuous deriv

Yes it is true.